How many people in each district of Ulaanbaatar live within walking distance of a park?

This project measures what share of residents in each Ulaanbaatar district live within 500m of a park, using OpenStreetMap and WorldPop data in QGIS. QGIS is a geographical information system software application used to view, edit, print, and analyze spatial and geographical data. From my analysis, some districts had far less parks within a walking distance (500m) than others. Notably, only about 2.6% of Songinokhairkhan (Сонгинохайрхан) district residents is within a 500m range of a park. While, approximately 42.4% of city centre district Bayangol's (Баянгол) residents have such access. Bayanzurkh (Баянзүрх) district is interesting due to it harboring the biggest park in the city, The National Park (Үндэсний Цэцэрлэгт Хүрээлэн), raising the number to an estimate of 38.8%.

Ulaanbaatar can often find itself at the top of most polluted cities in the world during winter times by IQAir rankings. This is largely due to mass unplanned urbanization from a nomad lifestyle and lack of infrastructure in ger districts. So, it is essential for green spaces to coexist within the city as trees absorb carbon dioxide (CO_2) and release oxygen (O_2) back into the atmosphere, allowing for a cleaner air condition. From an individual perspective, I am interested in studying to become an environmental engineer in the future as I have recently volunteered in an international event known as COP17.

Credits To:
-© OpenStreetMap contributors

-Bondarenko M., Priyatikanto R., Tejedor-Garavito N., Zhang W., McKeen T., Cunningham A., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A.. 2025 Constrained estimates of 2015-2030 total number of people per grid square at a resolution of 3 arc (approximately 100m at the equator) R2025A version v1. Global Demographic Data Project - Funded by The Bill and Melinda Gates Foundation (INV-045237). WorldPop - School of Geography and Environmental Science, University of Southampton. DOI:10.5258/SOTON/WP00839

-QGIS version 4.2.3-Belém do Pará

QGIS is a program that stacks layers that includes tools that combine them.
(Tools used for each step is provided in bold letters below)

Phase 1: Setting Up

1. **Set The Project CRS (ESPG: 32648)**. A CRS (coordinate reference system) is how the program turns the curved earth into flat coordinates. This specific one measures in meters, which makes certain tools accurate in its measurements.
2. **Download data with QuickOSM.** OpenStreetMap is a free map database, and QuickOSM pulls out specific things by tag, such as leisure=park. This gives layers such as districts, parks, gardens and residential areas. Each one becomes a vector layer: shapes (polygons) with a table of information attached.
3. **Reproject the layers.** Each layer has its own CRS. I converted them to the meter-based one so they line up and measure correctly.

Phase 2: Cleaning

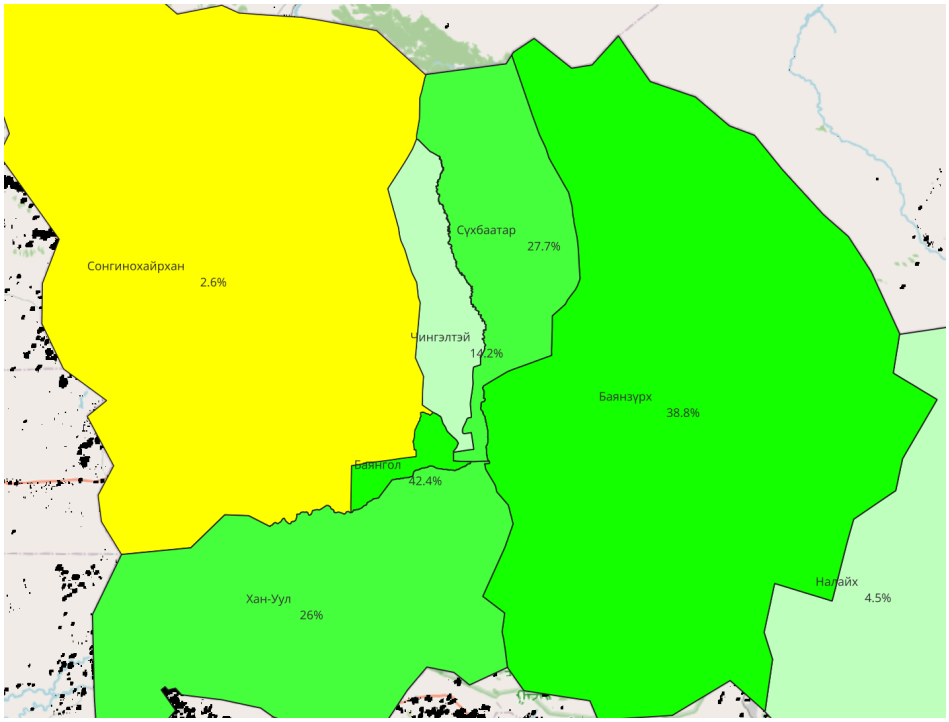
4. **Fix Geometries.** Some OSM shapes are broken (for example, edges that cross themselves). Broken shapes make later tools fail, so I repaired them first.
5. **Field Calculator (area_ha).** Every layer has an attribute table, a spreadsheet behind the shapes. I added a column calculating each park's area, then used Select by Expression to remove tiny ones.

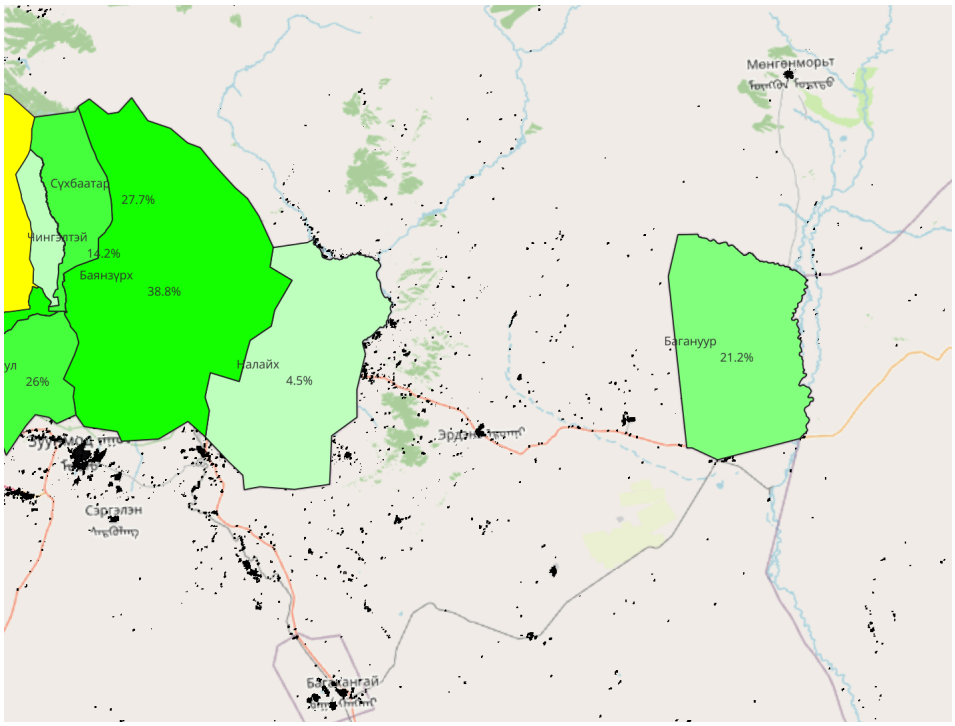
Phase 3: Analysis

6. **Buffer (500 m and 1000 m).** This draws a zone around each park. Anywhere inside it is within walking distance. "Dissolve" merged overlapping zones into one shape.
7. **Download the WorldPop raster.** This is a raster layer: a grid of cells where each cell holds a number, here the estimated number of people in that cell. Vectors are shapes, rasters are grids.
8. **Clip Raster by Mask Layer.** I cut the raster down to your study area, and then again to just the 500 m park zone. The second one only contains people who live near a park.
9. **Difference.** This subtracts one layer from another. Residential areas minus the park zone leaves the red areas, the neighborhoods that are too far from a park.
10. **Zonal Statistics (twice).** This adds up the raster cells inside each district's shape. Run on the full raster, it gave a total population per district. Run on the clipped one, it gave the population near parks.
11. **Field Calculator (pct_500).** I divided the two numbers to get the percentage of each district's residents within 500 m of a park. This gave me the actual result.

Phase 4: Showing and saving

12. **Graduated symbology.** This colors each district by its pct_500 value, which turns a table of numbers into a map.
13. **Export as GeoPackage.** This saves the layer's data to a file that lasts, rather than a temporary layer that disappears.





The percentage of residents of districts with the most access to a walking distance of a park (500m.):

1. Bayangol (Баянгол)-42.4%
2. Bayanzurkh (Баянзүрх)-38.8%
3. Sukhbaatar (Сүхбаатар)-27.7%
4. Khan-Uul (Хан-Уул)-26%
5. Baganuur (Багануур)-21.1%

6. Chingeltei (Чингэлтэй)-14.2%
7. Nalaikh (Налайх)-4.2%
8. Songinokhairkhan (Сонгинохайрхан)-2.6%

The districts with the worst percentage of access to local parks are often those with the most ger districts, which points to the idea of ger districts still lacking proper funding and infrastructure, while the central districts remain the leading percentage due to better city infrastructure.

This method of calculating percentages within this program has its limitations, such as:

- OSM may be incomplete, so low values can partly reflect missing parks.
- A straight-line 500 m buffer ignores real walking routes, rivers and roads.
- WorldPop is a modeled estimate, not a census count.
- The 0.1 ha cutoff was a choice, and a different cutoff would change the numbers.

This project utilizes QGIS, OpenStreetMap, and WorldPop data to calculate the percentage of residents within a 500m walking distance of a park across Ulaanbaatar's districts. The analysis reveals a stark contrast in green space access, with central districts like Bayangol achieving 42.4% accessibility compared to just 2.6% in the ger district of Songinokhairkhan. Despite methodological limitations such as straight-line buffering and incomplete mapping data, the findings highlight significant infrastructure disparities across the city. Of course, this research could be further improved by the inclusion of other sets of data or use of plugins with the software.